

(m, k) -code

↳ k = nr. of digits of the message
↳ nr. of digits of the code word (what we encoded)

• 2^k = nr. of messages to transmit

• We work in $\mathbb{Z}_2 = \{\hat{0}, \hat{1}\}$ with vectors $(x_1, x_2, x_3, \dots)$

↓
skip the hats

• ex: $(1, 0, 1) \rightarrow 101$ (not binary)

$\varphi: \mathbb{Z}_2^k \rightarrow \mathbb{Z}_2^m$ linear map

$[\varphi]_{EE'} = G$ = the generator matrix

↓ ↳ canonical basis of $\mathbb{Z}_2^m$.

canonical basis of $\mathbb{Z}_2^k$

$G = \begin{bmatrix} \textcircled{P} \\ \vdots \\ \textcircled{P} \\ \vdots \\ I_k \end{bmatrix} \Rightarrow H = \begin{bmatrix} I_{m-k} & \textcircled{P} \end{bmatrix}$

↓ generator matrix ↓ parity check matrix

If $H \cdot [u] = 0 \Rightarrow [u]$ is a code word.

Minimum Hamming distance = d

↳ the mim. nr. of columns of H which I can add to get 0. vector!
↓

• A code can detect t number of errors, where $\boxed{t+1 \leq d}$

• A code can correct t number of errors, where $\boxed{2t+1 \leq d}$

* The Cryptosystem Kettle

* King's poisoned wine

②. $f = 1 + x^3 + x^4 + x^7 \in \mathbb{Z}_2[x]$
 $g = x + x^2 + x^3 + x^6 \in \mathbb{Z}_2[x]$

$(8, 4)$ -code generated by a polynomial $p = 1 + x^2 + x^3 + x^4 \in \mathbb{Z}_2[x]$.

f, g code words?

$m = 8 \Rightarrow$ code words of 8 digits

$k = 4 \Rightarrow$ messages of 4 digits

degree(f) $< m$ if true $\Rightarrow$ may be code word.

degree (g) < m if true $\Rightarrow$

$$f \div p$$

$$\begin{array}{r} x^7 + x^4 + x^3 + 1 \mid x^4 + x^3 + x^2 + 1 \\ x^7 + x^6 + x^5 + x^3 \\ \hline x^6 + x^5 + x^4 + 1 \\ x^6 + x^5 + x^4 + x^2 \\ \hline \end{array}$$

$x^2 + 1 \neq 0 \Rightarrow f$ not a code word.
residue

$$\begin{array}{r} x^6 + x^3 + x^2 + x \mid x^4 + x^3 + x^2 + 1 \\ x^6 + x^5 + x^4 + x^2 \\ \hline x^5 + x^4 + x^3 + x \\ x^5 + x^4 + x^3 + x \\ \hline 0 \\ \parallel \\ g \text{ code word.} \end{array}$$

③ (6,3) - code generated by $p = 1 + x^2 + x^3 \in \mathbb{Z}_2[x]$

Write all code words.

messages of $k=3$ digits

$$2^k = 2^3 \text{ messages} \Rightarrow M = \{000, 001, 010, 011, 100, 101, 110, 111\}$$

011 $\Rightarrow m = 0 \cdot x^0 + 1 \cdot x^1 + 1 \cdot x^2 = x + x^2$

$$u \cdot x^{m-k} = (x + x^2) \cdot x^3 = x^5 + x^4$$

$$r = u \cdot x^{m-k} \pmod{p} = x^5 + x^4 \pmod{1 + x^2 + x^3} = x^2$$

$$\begin{array}{r} x^5 + x^4 \mid x^3 + x^2 + 1 \\ x^5 + x^4 + x^2 \\ \hline x^2 \end{array}$$

$$\text{CODE WORD } r = u \cdot x^{m-k} + r = x^2 + x^4 + x^5$$

$$\Rightarrow \underline{0} \underline{0} \underline{1} \underline{0} \underline{1} \underline{1}$$

(Same for the rest)

④ $G = \begin{bmatrix} -P \\ I_3 \end{bmatrix} \in \mathcal{M}_{6,3}(\mathbb{Z}_2)$

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Parity check matrix (H)? $m=5$

All code words? $k=3$

$$H = [I_{m-k} : P] = [I_2 : P] = \begin{bmatrix} 1 & 0 & : & 0 & 0 & 1 \\ 0 & 1 & : & 1 & 1 & 1 \end{bmatrix}$$

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$H \cdot [u] = 0 \Leftrightarrow H \cdot \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} u_1 + u_5 = 0 \Rightarrow u_1 = -u_5 \Rightarrow u_1 = u_5 \\ u_2 + u_3 + u_4 + u_5 = 0 \Rightarrow u_2 = u_3 + u_4 + u_5 \end{cases}$$

Code words: $\{(u_5, u_3 + u_4 + u_5, u_3, u_4, u_5) \mid u_3, u_4, u_5 \in \mathbb{Z}_2\}$.

ex: $(1, 1, 1, 1, 1), (0, 0, 1, 1, 0), (0, 0, 0, 0, 0)$
Trivial!

$$\textcircled{5} \quad G = \begin{bmatrix} P \\ I_4 \end{bmatrix} \in \mathcal{M}_{9,4}(\mathbb{Z}_2)$$

$$n=9, k=4$$

$$P = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

min. Hamming dist.?

discuss $\begin{cases} \text{error-detecting} \\ \text{error-correcting} \end{cases}$

$$d=3$$

$$C_1 + C_3 + C_8 \Rightarrow 0$$

$$H = [I_{m-k} : P] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow$ min. Hamming distance $= 3 = d$.

• detects t errors, $\boxed{t+1 \leq d}$

$$t+1 \leq 3$$

$$t \leq 2 \Rightarrow t \text{ at most } 2.$$

• corrects t errors, $\boxed{2t+1 \leq d}$

$$2t+1 \leq 3$$

$$t \leq 1 \Rightarrow t \text{ at most } 1.$$

! equality happens in special cases (usually, it is strict inequality)

⑥ (9,4) - code

using G from (ex 5), encode: 1101, 0111, 0000, 1000.

$G = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$1101 = e_1 + e_2 + e_4$$

$$\begin{aligned} g(1101) &= g(e_1 + e_2 + e_4) \xrightarrow{\text{lin. map.}} g(e_1) + g(e_2) + g(e_4) = \\ &= (001011000) + (010100100) + (011010001) = \underbrace{(000101101)}_{n=9 \text{ digit code word.}} \end{aligned}$$

$$0111 = e_2 + e_3 + e_4 \rightarrow \mathcal{E}(0111) = (100110111)$$

$$0000 \rightarrow 8(0000) = (0000000000)$$

$$\underline{1000} \rightarrow 8(1000) = (001011000)$$

⑧ (7,3)-code generated by $p = 1 + x^2 + x^3 + x^4 \in \mathbb{Z}_2[x]$.

Generator matrix?

Parity check matrix?

$$\gamma: \underbrace{\mathbb{U}_2^3}_E \rightarrow \underbrace{\mathbb{U}_2^7}_{E'}$$

$$[\delta]_{EE'} = G.$$

$$E = (e_1, e_2, e_3)$$

E' not interested.

$$\underline{e_1 = 100} = 1 \cdot x^0 + 0 \cdot x^1 + 0 \cdot x^2 = \underline{1 = m}.$$

$$\text{Ne} \cdot X^4 = X^4$$

$$h = x^4 \pmod{p} = x^3 + x^2 + 1.$$

$$N = x^4 + x^3 + x^2 + 1$$

$$\Rightarrow \underbrace{1 \ 0 \ 1 \ 1 \ 1 \ 0 \ 0}_{7 \text{ digit}} = 8(e_1)$$

7 digit

$$\underline{e_2 = 010} \Rightarrow \underline{m = x}$$

$$\underline{m \cdot x^4 = x^5}$$

$$\underline{r = 1}$$

$$\underline{r = x^5 + x^2 + x + 1 = x^8(e_2)}$$

$$\Rightarrow \underline{1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0}$$

$$\underline{e_3 = 001} \Rightarrow m = \dots x^2$$

$$m \cdot x^4 = x^6$$

$$r = x^3 + x^2 + 1$$

$$r = x^6 + x^3 + x^2 + 1$$

$$\Rightarrow \underline{1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1} = x^8(e_3)$$

